Implementation details for Full_Run_Fin.mlx program

Section 1 Constants and Setup

- Loads droneArmMaterials.mat file
- Defines base drone specifications/assumptions from instructions: frame/electronics mass (1.0 kg), quadcopter arms (4).
- Establishes operational criteria target Thrust-to-Weight Ratio (TWR = 2.0).
- Establishes and generates geometric, FEA, and mesh models

Section 2 Display models (For Devs)

- A section to view the models the program creates in order to check which faces to lock and apply forces to

Section 3 Volume, Thrust to weight ratio, and cost calculations

- 2 volumes are calculated
  - Vol# using the volume function for the drone arm
    - This is used for everything else besides costs. Allowing the program to calculate the mass of the arms
  - Vol#c using the maximum value in each axis to calculate the volume of a block
    - This is to give more accurate cost estimates as you would have to cut from a block of material in order to make the arm.

        - Model 2's is hand calced due to its stl file not being compatible with the other method of extracting the maximum value for each axis
- 2 cost variables are calculated (This is due to the confusion regarding the cost variable given by the droneArmMaterials.mat file. As the cost was $/m which was confusing as the team was more use to things like $/m^3 or $/Kg)
    - cost# uses the cost as ($/m) for calculations
        - This assumes that the units given were correct. So, that variable gets cubed in order to calculate the cost of the arms
    - cost#g uses the cost as ($/Kg) for calculations
        - This assumes that the units given were incorrect. So, that variable gets multiplied by mass of the arm instead to calculate the cost of the arms
- For loop (Run calculations for vol, costs, max payload, and "Thrust to Weight Ratio")
    - Please refer to earlier cost and vol sections for specifics of the calcs
    - The maximum possible payload is also calculated here.
    - The "Thrust to Weight Ratio" is technically calculated here due to the fact that the TWR is already locked to 2:1 or 2. So, if a payload is less than the maximum possible, then the TWR is higher than 2:1 or 2.

Section 4 Applying and loading forces

- Applies and locks selected faces. (Reason for section 2)
- An equation is placed into the the surface traction being F/SA
    - F = 9.07N refer to code for why (Open code or look at its pdf)

    - SA is handcalced for each model. Values were tuned to SolidWorks as a reference in FEA

Section 5 FEA Calculations

- Uses a switch case organize and fulfill FEA calculations
    - Store FEA results in variable for later visualization
        - Formatted "model_#_(material name)"
    - Finds max stress and displacement for each material for every arm.
    - Calculates the factor of safety for each material for every arm.

Section 6 Arm Viability Check

- This section quickly checks whether the minimum requirements are met by the drone arm with the selected material
    - **Max Payload weight >= 0.5Kg** As specified by the problem documentation
    - **FoS >= 1.5** As suggested by the problem documentation
    - **Max displacement <= 0.001m or 1mm** Value we set for ourselves
- This allows us to quickly check that conditions are met by the arm

Section 7 Setting up Tables

- This section creates 4 tables and the next non-numbered section displays said tables with the model/arm it is referring to.

Section 8 Visualizer

- This section uses the “Visualize PDE Results” task to allow for easier viewing of the model(s)/arm(s) stress and displacement visualizations.